

WEEK 1 – SUPERSTORE SALES DATA IMPORT, POWER QUERY & DATA CLEANING

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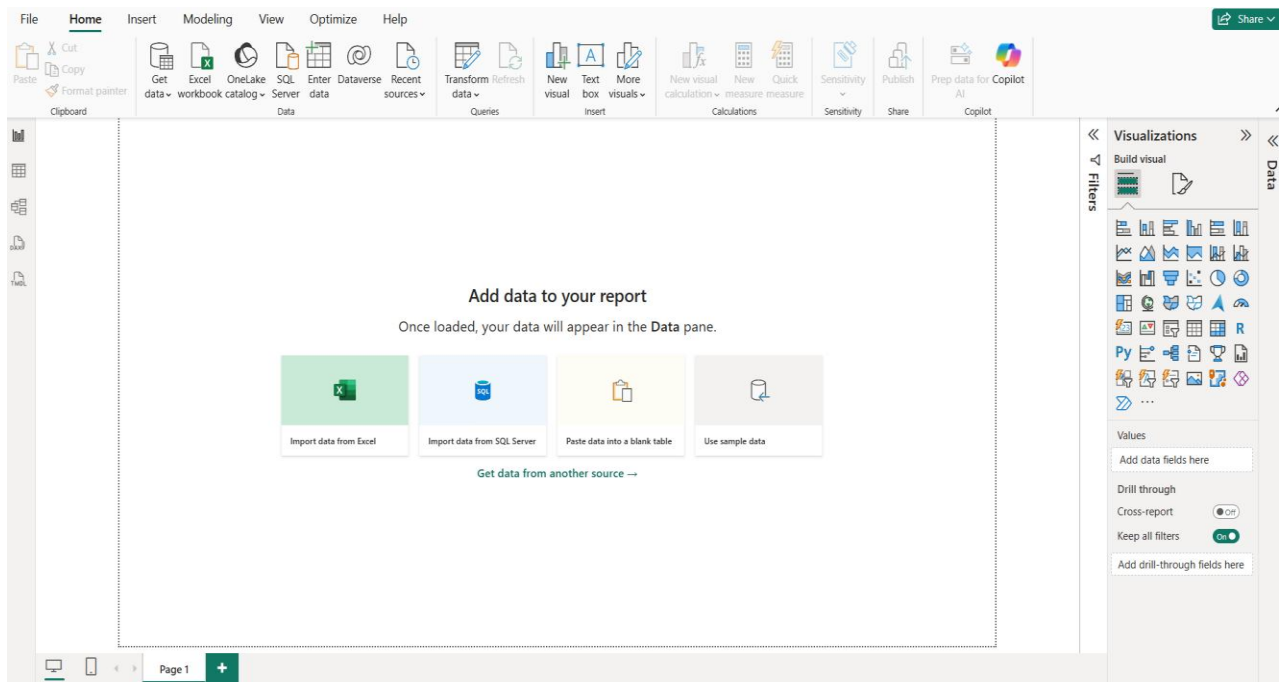
INTRODUCTION:

Microsoft Power BI is a data visualization and business intelligence tool used to import, clean, transform, analyze, and visualize data.

In this project, the Superstore Sales dataset was used to understand the basic data preparation process in Power BI. The dataset was first imported into Power BI and then cleaned using Power Query Editor.

The main activities performed during this project were:

- Importing the dataset
- Opening Power Query Editor
- Formatting date columns
- Checking data quality
- Identifying duplicate records
- Creating additional date columns
- Applying the transformations



Objective

The main objective of this project is to prepare the Superstore dataset for further analysis and visualization.

Objectives of the Activity

1. Import the dataset into Power BI.
2. Understand the Power Query environment.
3. Convert date fields into the correct format.
4. Check the quality of the available data.
5. Remove duplicate records where required.
6. Create useful date-related columns.
7. Apply all transformations to the Power BI model.

Proper data preparation helps in creating accurate reports and meaningful visualizations.

IMPORT THE DATASET:

The Superstore dataset was imported into Power BI using the Text/CSV data source option.

After selecting the dataset file, the available data preview was displayed. The Transform Data option was selected so that the dataset could be cleaned before being used for visualization.

Steps

Home → Get Data → Text/CSV → Select Dataset → Transform Data

Purpose

Importing the dataset allows Power BI to access the raw information and prepare it for analysis.

Result

The Superstore dataset was successfully loaded into Power Query Editor.

File

Home

Transform

Add Column

View

Tools

Help

Close & Apply *

New Source * Sources *

Recent

Enter Data

Data source

Data Sources

Manage Parameters *

Parameters

Export query settings

Output Data

Advanced Editor

Refresh Preview

Manage *

Choose Columns *

Remove Columns *

Manage Columns

Keep Results

Remove Rows *

Reduce Rows

Split Column *

Group By

Sort

Data Type: Whole Number *

Use First Row as Headers *

Replace Values

Merge Queries *

Append Queries *

Combine Files

Combine

Queries [1]

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Table.TransformColumnTypes(*Promoted Headers*,{{"Row ID", Int64.Type}, {"Order ID", type text}, {"Order Date", type text}, {"Ship Date",

samplesuperstore

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POWER QUERY EDITOR:

Power Query Editor is used to clean and transform data before creating reports.

It provides several options for modifying the dataset, such as:

- Changing data types
- Removing rows
- Removing duplicates
- Creating new columns
- Checking data quality

- Filtering data

The imported Superstore data was displayed in table format inside Power Query Editor.

Result

The dataset was ready for the required cleaning and transformation operations.

ORDER DATE & SHIP DATE FORMATTING:

The Order Date column was checked and converted into the appropriate date format.

The Using Locale option was used to ensure that Power BI correctly interpreted the date values.

Steps

Transform → Data Type → Using Locale → Date → Select Using Locale → OK

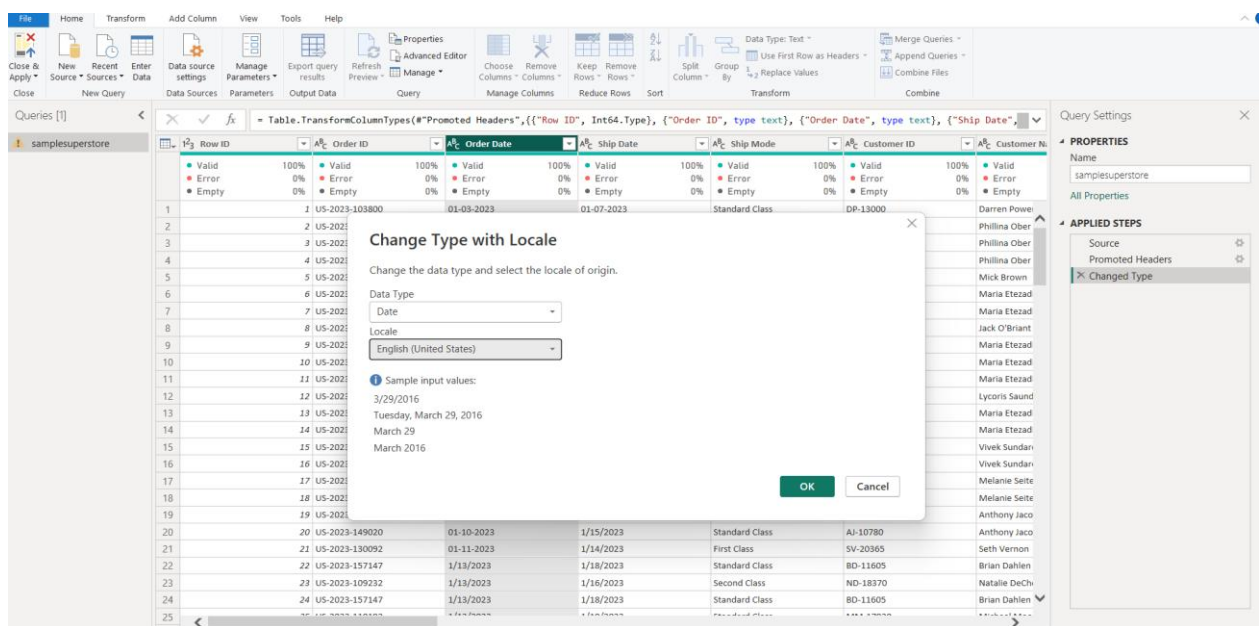
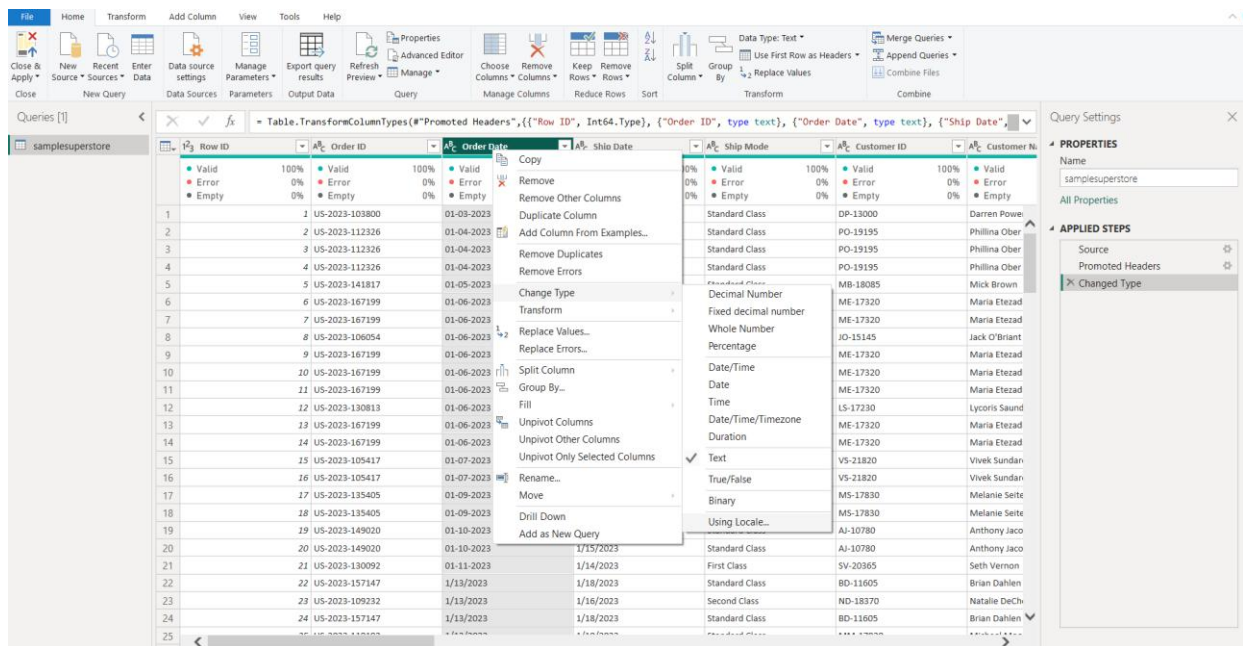
Why This Step Was Performed

Correct date formatting is important because date fields are later used for:

- Year analysis
- Monthly analysis
- Day analysis
- Time-based visualizations

Result

The Order Date column was successfully converted into the required date format.



The **Ship Date** column was also converted into the correct date format.

The same **Using Locale** method was followed to make sure that the shipping date values were properly recognized.

Steps

Transform → **Data Type** → **Using Locale** → **Date** → **Select Locale** → **OK**

Purpose

Having both Order Date and Ship Date in the correct format makes it easier to perform date-based analysis later.

Result

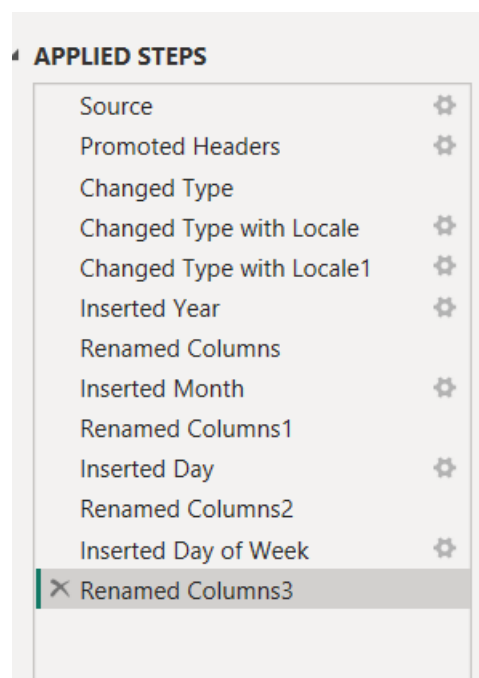
The Ship Date column was successfully formatted.

CREATING COLUMNS:

After creating the required temporal columns, the dataset was reviewed in Power Query Editor. The newly generated fields were checked to make sure that the values corresponded correctly with the original Order Date.

The temporal transformation process created useful fields such as Order Year, Order Quarter, Order Month, Month Number, Order Day, Order Day of Week, Day Number, and Order Week. These fields provide different levels of date information and make the Superstore dataset more suitable for time-based analysis.

The Applied Steps panel was also reviewed to verify the sequence of transformations performed during the data preparation process. The source report explains that this panel records the transformations and helps verify that the preparation process was completed correctly.



DATA TRIM AND CLEANING:

The text data was cleaned in Power Query Editor to improve consistency. The Trim function removed unnecessary spaces, while the Clean function removed unwanted characters from the selected text fields.

Trim

1. Select the required text column.
2. Go to Transform → Format → Trim.
3. Apply the transformation to the required columns.
4. Review the values after trimming.

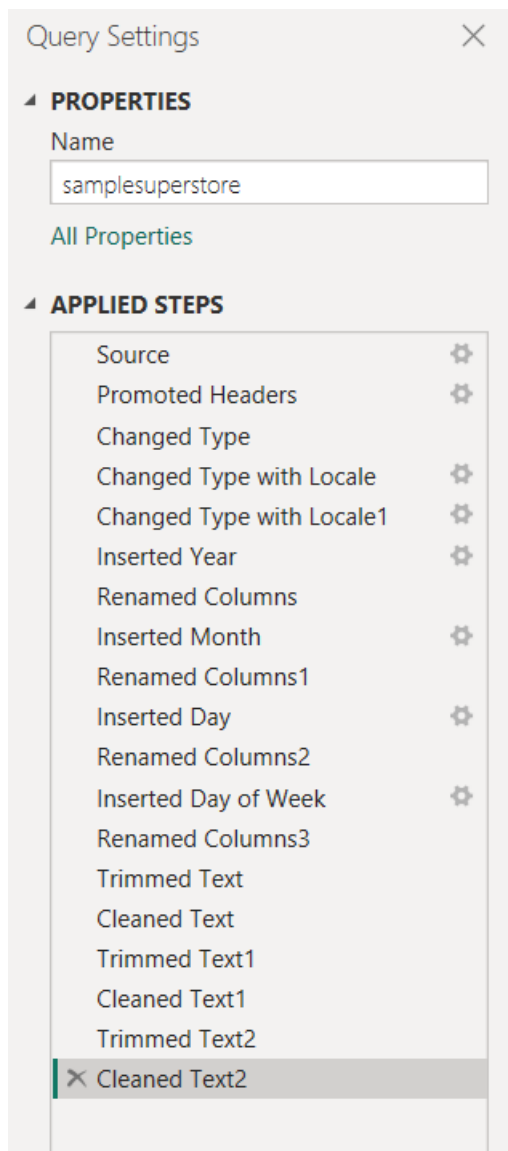
Result: Extra spaces were removed, making the text values more consistent.

Clean

1. Select the required text column.
2. Go to Transform → Format → Clean.
3. Apply the transformation to the required columns.
4. Check the updated values.

Result:

Unwanted characters were removed, producing cleaner and standardized text data.



CONCLUSION:

The Superstore dataset was successfully prepared using Microsoft Power BI and Power Query Editor.

During this activity, the dataset was imported and examined before performing the required transformations. The Order Date and Ship Date fields were converted into appropriate date formats. Data quality was checked using Column Quality, and duplicate records were handled.

Additional date-related columns such as Year and Name of Day were also created. Finally, all transformations were applied using Close & Apply.

The prepared dataset can now be used for the next stage of the project, which involves creating charts, KPI cards, slicers, and other Power BI visualizations.